

Data scientist with **over a decade of research and production experience** in probabilistic, multi-objective and generative AI systems. Currently leading generative-AI and machine-learning work for enterprise wireless networks at Extreme Networks, where language-model reasoning is grounded in statistical signal processing and domain knowledge graphs to **cut network investigation time from hours to seconds**. Doctorate in AI, publications at AAAI, IJCAI, UAI, CP and ICTAI, **nine granted US patents**.

10+

YEARS IN AI / ML

9

GRANTED US PATENTS

7

PUBLICATIONS

PhD

AI, NUI CORK

IEEE

SENIOR MEMBER 2026

CORE EXPERTISE

Detailed practice areas at abdulrazakucc.github.io/work.html

- **Generative AI for operations.** Grounded LLM reasoning, autonomous agent workflows, retrieval and knowledge graphs, multi-modal telemetry correlation.
- **Multi-objective optimisation.** Preference elicitation and inference, Pareto optimisation, constraint programming under imprecise trade-offs.
- **Probabilistic modelling.** Bayesian inference, probabilistic graphical models, influence diagrams, calibrated uncertainty for industrial decisions.
- **Research-to-production engineering.** Real-time streaming and batch ML pipelines on AWS and Azure, from prototype to service-level objectives.

PROFESSIONAL EXPERIENCE

Staff Data Scientist, [Extreme Networks](#)

2020 – PRESENT • SALEM, US (PREV. CORK, IE)

- Lead the data-science strategy for **generative AI in enterprise wireless networking**, combining language-model reasoning with statistical signal processing and a knowledge graph of devices, sites, topology and configuration.
- Designed **autonomous agent workflows** that investigate client behaviour, roaming failures and device-performance anomalies across thousands of access points and switches, and propose remediation plans to operators.
- Built **network-aware inference pipelines** and multi-modal correlation across telemetry, device health and client-experience metrics, suppressing false alerts and cutting investigation time on common issues from hours to seconds.
- Apply preference inference and multi-objective methods to **personalise networking recommendations** across widely differing deployment sizes.

Data Scientist, [Johnson Controls](#)

2017 – 2020 • CORK, IRELAND

- Led machine-learning work across the **intrusion-alarm, access-control and retail-analytics** product lines, delivering supervised, unsupervised and probabilistic models into commercial building-security products.
- Co-invented **eight granted US patents** on Pareto-optimal alarm classification, site-independent signature generation and rules-based misuse detection.
- Applied relational probabilistic programming (BLOG) and Bayesian inference (PyMC3) to alarm classification and event root-cause analysis across large multi-site estates.

SFI Post-Doctoral Fellow, [Insight Centre for Data Analytics, NUI Cork](#)

2014 – 2017 • CORK, IRELAND

- Held a **Science Foundation Ireland Post-Doctoral Fellowship** in a United Technology Research Centre programme on probabilistic graphical models, clustering and influence diagrams for industrial decisions.
- Extended the doctoral line to **lexicographic and weighted-average preference models** and interactive elicitation with IBM Research Labs Dublin; published at AAAI 2017, IJCAI 2015 and ICTAI 2015.

Doctoral Researcher, [Cork Constraint Computation Centre, NUI Cork](#)

2010 – 2014 • CORK, IRELAND

Funded by the **IRCSET / IBM Enterprise Partnership Scheme** within IBM's Risk Management Colloboratory. Developed exact and approximate algorithms for **multi-objective constraint optimisation and influence diagrams with imprecise trade-offs**, including interactive elicitation that minimises the input required from a decision maker.

Lecturer in Mathematics, [Osmania University](#)

2005 – 2010 • HYDERABAD, INDIA

Taught algebra, real and numerical analysis, calculus, probability and statistics to undergraduate cohorts.

SELECTED PUBLICATIONS

Full list: </research.html>

- AAAI**
2017 **Multi-objective Influence Diagrams with Possibly Optimal Policies**
R. Marinescu, A. Razak, N. Wilson • 31st AAAI Conference on Artificial Intelligence
- ICTAI**
2015 **The Comparison of Multi-Objective Preference Inference Based on Lexicographic and Weighted Average Models**
Anne-Marie, A. Razak, N. Wilson • 27th IEEE International Conference on Tools with Artificial Intelligence
- IJCAI**
2015 **Computing Possibly Optimal Solutions for Multi-Objective Constraint Optimisation with Trade-offs**
N. Wilson, A. Razak, R. Marinescu • 24th International Joint Conference on Artificial Intelligence
- AIGM**
2015 **Multi-Objective Constraint Optimisation with Lexicographic Preference Models**
Anne-Marie, A. Razak, N. Wilson • Workshop on Algorithmic Issues for Inference in Graphical Models
- CP**
2013 **Multi-objective Constraint Optimisation with Trade-offs**
R. Marinescu, A. Razak, N. Wilson • 19th International Conference on Principles and Practice of Constraint Programming
- UAI**
2012 **Multi-objective Influence Diagrams with Trade-offs**
R. Marinescu, A. Razak, N. Wilson • 28th Conference on Uncertainty in Artificial Intelligence

GRANTED US PATENTS

Nine granted, 2019–2021 • </patents.html>

- Decision intelligence and maintenance reasoning** 3 PATENTS
- 2019 Interactive method to reduce the amount of trade-off information required from decision-maker in multi-attribute decision-making under uncertainty
- 2019 Virtual Maintenance Manager
- 2021 Root-cause analysis of event occurrences
- Alarm management and optimisation** 3 PATENTS
- 2020 Automated Alarm Panel Classification using Pareto Optimization
- 2021 Systems and methods for managing alarm data of multiple locations
- 2021 Building security system with alarm lists and alarm rule editing
- Building security analytics** 3 PATENTS
- 2020 Building security analysis system with site-independent signature generation for predictive security analysis
- 2020 Rules-based method of identifying misuse of emergency fire exits using data generated by a security alarm system
- 2020 Systems and Methods for Managing a Security System

TECHNICAL TOOLKIT

LANGUAGES & DATA

Python C++ (STL, Boost, Eigen)

SQL PySpark PyFlink

PLATFORMS

AWS SageMaker AWS Athena

Azure Databricks

Real-time streaming

MODELLING & SOLVERS

PyMC3 BLOG Swift PPL

ILOG CPLEX LPSOLVE

METHODS

LLM reasoning Knowledge graphs

Multi-modal AI Bayesian inference

Influence diagrams

Constraint programming

EDUCATION

PhD, Computer Science (Artificial Intelligence)

National University of Ireland, Cork • Cork
Constraint Computation Centre

NOV 2010 – MAY 2014

Thesis: Reasoning with Imprecise Trade-offs in Decision Making under Certainty and Uncertainty

MSc, Mathematics with Computer Science

Osmania University, Hyderabad, India
JUN 2003 – MAY 2005

BSc, Mathematics, Statistics and Computer Science

Osmania University, Hyderabad, India
JUN 2000 – MAR 2003

AWARDS & HONOURS

- IEEE Senior Member, elevated May 2026.
- Science Foundation Ireland Post-Doctoral Fellowship, Insight Centre for Data Analytics, NUI Cork (2014–2017).
- IRCSET / IBM Enterprise Partnership Scheme scholarship, funding the doctoral programme (2010–2013).
- Osmania University Merit Scholarship for the MSc programme (2003–2005).
- Ranked 40th of roughly 30,000 candidates in the Osmania University mathematics entrance test (2003).

ACADEMIC SERVICE

- Reviewer, **Computers and Electrical Engineering** (Elsevier) — 10+ manuscripts in AI, wireless networking and intelligent systems.
- Reviewer, **IEEE ICC** — AI submissions for the 2025 technical programme.
- Invited talks at **Sapienza University of Rome**, **NUI Galway**, **NUI Cork** and **IBM Research Labs Dublin**.
- Oral and poster presentations at AAAI, IJCAI, UAI, CP and ICTAI.

PROFILE

Nationality — Irish (2021), originally Indian

Languages — English, Hindi, Telugu, Urdu

Patents — Justia and Google Patents (inventor: Abdul Razak)

Research — Google Scholar, DBLP, ORCID, ResearchGate

Full detail, with context and links, at abdulrazakucc.github.io.